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How much should we pay for interconnecting electricity markets? A real options approach[☆]

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ABSTRACT

An interconnector is an asset that gives the owner the option to transmit electricity between two locations. In financial terms, the value of an interconnector is the same as a strip of real options written on the spread between power prices in two markets. We model the spread based on a seasonal trend, mean-reverting Gaussian process, and mean-reverting jump process and express the value of these real options in closed-form. The valuation tool is applied to five pairs of European neighboring markets to value a hypothetical one-year lease of the interconnector using different assumptions about the seasonal component of the spread, and different liquidity caps which proxy for the depth of the interconnected power markets. We derive no-arbitrage lower bounds for the value of the interconnector in terms of electricity futures contract and find that, depending on the depth of the market, the jumps in the spread can account for between 1% and 40% of the total value of the interconnector. The two markets where an interconnector would be most (resp. least) valuable are Germany and The Netherlands (resp. France and Germany). Finally, we provide rules of thumb to interpret the different interconnector values.

1. Introduction

Electricity markets have undergone a series of fundamental changes sparked by the liberalization of this industry. The first stage of liberalization required privatization of all or most of the generation assets, as well as privatization of the transmission grid which transports electricity from the generation points to the end consumer. Another important step in the development of the wholesale electricity markets is to exploit price differentials between locations by building interconnectors which are bi-directional transmission lines connecting the grids of two locations or the grids of two countries. Although interconnecting different grids is at the top of the political agenda in many countries, the decision to build them depends on their financial value.[1]

Electricity prices are characterized by exhibiting extreme volatility and by undergoing abrupt changes (large upward spikes and large downward jumps), as well as fast mean reversion to a seasonal trend. This extreme behavior is also present in the difference between prices of two locations and explains why interconnecting two markets could be profitable. The main question we address in this paper is how to value an interconnector. One of the key features that drives the financial value of an interconnector is that the owner has the right, but not the obligation, to transmit electricity between two locations. Therefore, once it has been built, the financial value of an interconnector is given by a series of real options which are written on the price differential between two electricity markets.

In this paper we propose a valuation tool that uses real options theory to consider the problem and we employ market data of five pairs of European neighboring countries to value hypothetical interconnectors under realistic assumptions. The value of an interconnector is given by a strip of European-style options (Bull Call Spreads) written on the spread between the two markets and the valuation formula is in closed-form and is quick to implement. Our model for the spread captures the main characteristics of the dynamics of price differentials: jumps in both directions, high seasonal volatility, and fast mean reversion to a seasonal trend. We propose an algorithm to detect jumps where the emphasis is placed on avoiding misclassifying mean reversion as jumps. We estimate the parameters of the spread model and find that the introduction of jumps in the model delivers gains in the in-sample performance of

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[1] For example, see Department of Energy (2002) and European Commission (2008) for the policy steps towards interconnecting grids in the US and European Union.

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between 20% and 48% with respect to a misspecified or "naive" modeling which jumps are not included.

We show valuations under different liquidity caps, which proxy for the depth of the interconnected power markets. We also derive no-arbitrage lower bounds for the value of the interconnector in terms of electricity futures contracts of the respective power markets. We find that most of the time these bounds are satisfied, but there are days where the value of the interconnector is given by the no-arbitrage bound instead of the price delivered by the sum of the prices of real options. We find that, depending on the depth of the market, the jumps in the spread can account for between 1% and 40% of the total value of the interconnector. The two markets where an interconnector would be most (resp. least) valuable are Germany and The Netherlands (resp. France and Germany). The markets where off-peak transmission between the two countries is more valuable than transmission during peak times are: France and Germany, France and UK, and The Netherlands and UK. We also provide "rules of thumb" to summarize the different drivers of the interconnector value.

The rest of this paper is organized as follows. Section 2 reviews the literature on real options in commodity markets, electricity price models, and spread options. Section 3 discusses the data and why interconnectors are valuable. Section 4 frames the financial value of interconnector leases as a strip of European Bull Call Spread options and Section 5 derives no-arbitrage bounds for the lease based on traded assets such as electricity forwards and futures. Section 6 presents a model for the spread and derives valuation formulae. Section 7 describes how the model parameters are estimated and describes the algorithm that we propose to detect jumps and disentangle mean-reversion from jumps. Section 8 shows values of a one-year interconnection lease for five pairs of European neighboring markets and discusses the no-arbitrage bounds using futures data. Section 9 concludes.

2. Literature review

In energy markets there are many projects whose value depends on the flexibility of being able to delay decision-making until more information becomes available. These decisions can include delaying or accelerating production, postponing entry, scaling production, changing technology, etc., see Trigeorgis (1996), Brennan and Trigeorgis (2000), and Keppo and Lu (2003). In many cases the flexibility embedded in some types of project is what drives most of their value. For example, some electricity plants are only economically viable to operate when market prices are very high, otherwise they must be "switched off". Moreover, gas-fired plants are very valuable because relative to other plants (for instance nuclear and coal-fired ones) it is easier to ramp up or ramp down according to the level of market prices. Neglecting these embedded real options may seriously undervalue some projects to the extent that they might seem to deliver a negative NPV when in fact they are viable.

In the natural gas and liquified natural gas (LNG) industry, the value of some assets and financial instruments principally depends on the flexibility that these assets provide to their management. For example, the market value of a natural gas storage facility depends on the ability to store gas during times of low prices, and the ability to bring the stored gas to market at times of high prices, see Chen and Forsyth (2007), Boogert and De Jong (2008), and Carmona and Ludkovski (2010). The value of natural gas supply contracts depends on the flexibility of the shipper to interrupt delivery during the life of the contract, see Jaillet et al. (2004), and Cartea and Williams (2008).

Real options in electricity markets are also key components in project valuation. Power plants that offer operational flexibility derive most of their value from the option to produce electricity when prices are high. These options are valuable because wholesale electricity prices are extremely volatile, but the extreme behavior of power prices makes electricity prices a difficult commodity to model. Modeling electricity interruptible supply contracts and electricity swing contracts has been undertaken by Hambly et al. (2009), Keppo (2004), and Kamat and Oren (2002). Modeling power prices, and other contracts such as futures and forwards, can be found in Borak and Weron (2008), Cartea and Figueroa (2005), Cartea and Villaplana (2008), Coulon and Howison (2009), and Escribano et al. (2011), Hikspoors and Jaimungal (2008), Kiesel et al. (2009), Pirrong and Jermakyan (2008), Roncoroni (2002), and Weron (2006).

The other literature that is relevant to our approach of valuing electricity interconnectors is that related to spread options in energy commodities: Dempster et al. (2008), Hikspoors and Jaimungal (2007), Benth and Saltyte-Benth (2006), Marckhoff and Muck (2009a, 2009b), and for a thorough and extensive survey on the topic see Carmona and Durrleman (2003).

3. The Market for interconnectors and data

An important feature common to all energy commodities is that their market value depends on the location and the date that the delivery of the commodity takes place. This is particularly important for electricity where date and location are crucial determinants of market clearing prices because electricity must be consumed immediately upon delivery, while consumption of other energy commodities such as gas and oil can be deferred by either postponing delivery or by storing them. In fact, as a consequence of the non-storability of electricity, one can think of electricity delivered over different intervals of the day, or throughout periods of the year, as different goods.[2]

A further consequence of not being able to store electricity is that, strictly speaking, there are no electricity spot prices as commonly understood. Market clearing prices must be agreed prior to delivery at a time when production and demand are not known for sure; this uncertainty is resolved at the time when the physical transaction occurs. In addition, for this market clearing process to function, it is necessary for the system operator to ensure that there is sufficient capacity in the grid to secure transmission from generators to both retailers and consumers. Therefore, the convention in the market and the literature is to treat the day-ahead prices as the spot prices, although their structure is more akin to that of a forward contract. Depending on the market one can find different day-ahead quotes (prices today for next-day delivery) for contracts that dispatch electricity over fixed-time intervals during the delivery day. For example, in the UK it is possible to individually trade each of the 48 half-hours 1 day prior to delivery, while in the Nord Pool it is possible to individually trade each of the 24 h 1 day prior to delivery. Another standard way in which blocks of electricity are bundled is peak and off-peak. Peak hours correspond to a fixed interval of hours for business days characterized by high electricity demand, normally between 8am and 8pm. Off-peak hours belong to the interval between the end of a peak block and the beginning of the next one, and include the 24 h of weekends' days and holidays. The day-ahead peak and off-peak contracts specify delivery of 1 MWh for every hour of their corresponding time interval.

The owner of the interconnector capacity needs to schedule the flows according to prevailing market prices and the transmission costs in the two interconnected locations. In practice these decisions are generally taken on the day-ahead market. Thus, we assume that the decision to use the interconnector to dispatch electricity from A to B, or vice versa, is based on the peak and off-peak market prices observed in the day-ahead market, net of transmission costs.[3] Therefore, every day the owner of the interconnector capacity faces various alternatives. To commit to dispatching electricity the following day from A to B, or from

[2] Due to its non-storability, electricity is considered a non-traded asset, see Schwartz (1997).

[3] In spite of that, inefficient arbitrage transmission can occur. Bunn and Zachmann (2010) have shown that, under the presence of a dominant generator in one location, there might be electricity flows from high to low price area, even while most players trade in the opposite (efficient arbitrage) direction.

Table 1
Summary statistics. Summary statistics (mean, standard deviation, minimum, maximum, in Euros/MWh, and skewness and kurtosis) of spot prices and spreads. *Corr* is the sample correlation coefficient between the spot prices used to calculate the spread, all are significant at the 5% level. We perform an Augmented Dickey–Fuller test (ADF) on the prices using 21 lags. The null hypothesis is a unit root with constant trend in the null hypothesis, whose critical values are equal to −3.49, −2.87 and −2.59 for 1%, 5% and 10% confidence levels, respectively. The unit root hypothesis is always rejected in favor of the mean reverting alternative at 1% significance. Normality in the spreads is rejected in all cases at a 1% level according to the Jarque–Bera statistic (*JB stat*). In this table, we only take into account working days. For the markets where we have weekend data, main statistics do not change significantly, neither do the results of the tests of normality nor the unit root tests.

Panel A: spot prices

	France		UK		Germany		Nord Pool		The Netherlands	
Period	26/11/2001–02/07/2009		16/06/2000–01/07/2009		01/01/2002–29/04/2009		01/01/2000–04/07/2009		03/01/2000–02/07/2009	
# Obs.	2776		2359		2676		3473		2479	
	Peak	Off-peak	Peak	Off-peak	Peak	Off-peak	Peak	Off-peak	Peak	Off-peak
Mean	52.75	32.82	50.09	44.33	48.39	32.06	34.97	31.08	58.77	27.32
Std. Dev.	33.10	16.18	30.32	25.34	30.58	21.74	14.45	13.56	41.76	14.53
Min	5.06	4.79	14.78	15.97	0.80	−342.24	5.71	3.7883	1.78	1.93
Max	551.41	123.10	265.28	214.05	543.72	293.82	119.32	112.42	944.61	119.97
Skew.	3.93	1.12	1.99	2.01	4.18	−0.91	1.31	1.35	6.74	1.82
Kurt.	40.65	3.79	8.67	8.86	45.09	71.78	6.18	6.57	105.36	7.29
ADF	−14.62	−7.42	−12.14	−6.62	−24.28	−40.72	−5.70	−4.16	−22.25	−9.58

Panel B: Spread prices

	France–Germany		France–UK		Nord Pool–Germany		Germany–Netherlands		The Netherlands–UK	
	Peak	Off-peak	Peak	Off-peak	Peak	Off-peak	Peak	Off-peak	Peak	Off-peak
Mean	0.27	−1.71	2.84	−9.75	−13.42	−0.98	−10.41	4.76	10.77	−15.56
Std Dev	17.88	19.03	25.57	18.15	27.00	21.50	40.75	18.80	39.53	17.76
Min	−222.09	−256.51	−169.94	−166.53	−489.02	−270.45	−901.42	−389.92	−169.49	−165.90
Max	305.56	396.85	419.41	286.81	94.07	393.42	323.67	263.47	915.88	36.61
Skew.	2.50	3.26	3.74	0.21	−4.59	3.24	−6.34	−2.37	8.79	−2.67
Kurt.	110.71	158.34	64.80	40.97	60.76	101.67	121.81	139.59	166.90	16.05
ADF	−49.48	−51.77	−28.78	−27.45	−30.21	−42.01	−34.39	−49.66	−24.46	−13.41
Corr.	0.848	0.561	0.677	0.671	0.470	0.330	0.400	0.527	0.441	0.721
JB stat. ($\times 10^5$)	9.6	19	3.1	1.6	3.3	9.6	14	18	21	16

B to A, during the peak and off-peak hours. To decide not to dispatch electricity in any direction during the peak and/or off-peak period.

3.1. Data

European energy markets are undergoing important changes in the way they function and in how integration between them is evolving. Bunn and Gianfreda (2010) employ electricity forward and spot data to show that the degree of market integration between the French, German, British, Dutch and Spanish markets is increasing. Here we look at five electricity markets: Powernext (France), UKPX (the United Kingdom), EEX (Germany and Austria), APX (The Netherlands) and Nord Pool (Norway, Sweden, Finland and Denmark). Table 1 summarizes the data we use in this paper. For all markets, peak and off-peak day-ahead prices for weekdays are available.[4]

Panel A in Table 1 shows statistics for peak and off-peak prices for these five markets. Panel B in Table 1 hints at why interconnecting neighboring markets might be desirable. If we assume that transmission costs are around 5 Euros/MWh and if the price paid for interconnector capacity is seen as a sunk cost, then from a mean price point of view it would be profitable to transmit electricity across the different locations. For example, by looking at the mean of the historical spreads it seems that in the off-peak segment of the day it would be profitable to use the interconnector between France and the UK, and between The Netherlands and the UK. Similarly, in the peak segment, electricity would flow from Germany to Nord Pool, from The Netherlands to Germany, and from the UK to The Netherlands.

As Table 1 shows, the correlation between the prices used to calculate the locational spreads in Panel B are in all cases significant and relatively high. Spreads range from a minimum of 0.33 for off-peak hours between Nord Pool and Germany, to a maximum of 0.85 for peak hours between France and Germany; two markets that are already partially interconnected.

We perform an Augmented Dickey–Fuller (ADF) test on the prices and spreads using 21 lags. Although the power of this type of test is sensitive to heteroscedasticity and outliers, statistics suggest the rejection of the unit root hypothesis in favor of mean-reverting alternatives in all cases. This pattern is even stronger for spread prices than for the price levels. The Jarque–Bera statistics show that spreads are far from being Gaussian. Moreover, spreads present a significant non-zero skewness and larger kurtosis than the prices.

Other important statistics shown in Panel B of Table 1 are the maxima and minima of the spreads. For example, the minimum peak spread between Germany and the Netherlands is −901 Euros/MWh. The maximum spread is between The Netherlands and UK at 915 Euros/MWh. Although these are the extreme cases observable in the data and although they are not frequent occurrences in these markets, it prompts a very important question. Will it be possible for the owner of the interconnector capacity to take simultaneous short and long positions in the two locations when the market is undergoing such remarkable price differentials? Although there seems to be insufficient public information about the depth of these markets, market participants agree that these represent situations where liquidity in at least one of the two locations is too thin. Consequently, it does not seem plausible to assume that the owner of the interconnector capacity will be able to take advantage of such extreme situations; something that will need to be taken into account when valuing the real option held by the owner of the interconnector capacity. We will return to this issue in Section 4 below.

[4] In addition, for France, UK, and Nord Pool, we have data for weekends. The definition of peak hours differs across markets. For example, peak hours for France are between 9am and 8pm, Germany from 7am to 7pm, UK from 8am to 8pm, and Nord Pool from 7am to 10pm.

4. Valuing interconnection capacity: a strip of real options

Writing contracts on the difference between two or more assets has a long tradition in commodity markets. In the exchanges, all of the commonly traded energy spread options have the difference between a linear combination of energy futures contract as the underlying. These standard spread option contracts are written on the difference of futures contract between: electricity and natural gas (the spark spread), electricity and coal (the dark spread), electricity and a fuel including emission allowance costs (the clean spread), crude oil and one of its derivative products (the crack spread), and others.[5]

We note that all energy spread options that are traded in exchanges have payoffs based on futures contract. Consequently, models proposed in the literature to price options on spreads are designed to capture the stylized features of the underlying futures. Compared to more traditional asset classes such as equity, modeling commodity futures is relatively more involved due to the fact that energy futures have delivery periods (which can range from 1 day to years) rather than spot or instantaneous delivery, see Benth and Koekebakker (2008), Fusai et al. (2008), Borak and Weron (2008), and Fusai and Roncoroni (2008).

Our objective is to price the optionality provided by an interconnector that can exploit the wholesale electricity spot price differential between two markets.[6] There are two crucial features that differentiate our problem from the more traditional spread options studied in the literature. First, for the owner of the interconnector capacity, the underlying "asset" of the real option is MWh of electricity and not futures or forwards written on electricity.

Second, the value of interconnection capacity between two locations, for instance locations A and B, is equivalent to holding a strip of European-style options. The decision to use the interconnector to dispatch or not to dispatch electricity in any direction, at peak and off-peak hours, is based on the day-ahead market. That is, every day the owner of the interconnector exercises the right to use the capacity to simultaneously buy electricity in market A, to sell the same quantity of electricity in market B, or vice versa. In other words, the owner of the capacity holds four daily European options: two options on the spread between A and B; and two options on the spread between B and A (one option for peak and the other for off-peak). Since each individual option is only for 1 day, we cannot cast the valuation problem in terms of futures contracts since the delivery period for these will be at least 1 month. Nevertheless the information provided by futures contracts can be used to determine no-arbitrage bounds for the European options on the spread; this is discussed in detail below in Section 5.

A further assumption we make is that the capacity of the interconnector is small relative to that of the markets it is connecting. This is the same as assuming that the presence of the interconnector does not alter the price dynamics in either market; a plausible assumption for the cases we study below. Although our model does not endogenize the impact that the interconnector might have on the spread dynamics, our framework allows us to analyze different scenarios and look at the sensitivity of the value of the interconnector to: price volatility; price spikes; speed of mean reversion; and liquidity constraints when two markets are interconnected.[7]

In Panel B of Table 1 we showed the maxima and minima of the spread for different locations and argued that in these extreme conditions markets were too thin; in Fig. 2 we can also appreciate some of the extreme prices in the spread. In at least one of the locations it does not seem plausible to take long or short positions at the prices that produced such large spreads. Here we assume that during times of extreme price deviations, the owner of the interconnector capacity can take positions in both markets but we limit the extent to which he can profit from the situation. We do this by capping the amount of profit that can be extracted from in-the-money options upon exercise, when valuing interconnection capacity. We denote the maximum spread level, at which it is feasible for the owner of the interconnector capacity to take positions in both locations, by M and for simplicity assume that this liquidity cap is the same regardless of whether it is an option on the peak or off-peak spread.

The valuation problem thus reduces to being able to price European capped options. For ease of presentation let us focus on the spread between A and B, which we denote $S^{A,B}(t)$, and assume that it is for peak electricity, without specifying the particular hour during the peak segment. Let $C_p^{A,B}(S^{A,B}, M, t;T, K^{A,B})$ denote the price of a European call at time t, written on the spread $S^{A,B}(t)$ during peak time, but capping the maximum value at $M>0$, and expiring at a future date T with strike price $K^{A,B}<M$. The pay-off of such option is given by $\max(\min\{S^{A,B}(T),\ M\}-K^{A,B},\ 0)$. The option gives the right to transmit 1 MWh of electricity, during a designated hour of the day, but for ease of notation we do not specify the particular hour of the day.[8] The strike price represents the transmission costs between locations A and B, and time T represents the time in future periods when the decision will be made whether to use the interconnector capacity. Then, the price of the call is given by

$$C_p^{A,B}\left(S^{A,B}, M, t; T, K^{A,B}\right) = e^{-\rho(T-t)}\mathbb{E}_t\left[\max\left(\min\left\{S^{A,B}(T), M\right\} - K^{A,B}, 0\right)\right] \tag{1}$$

where ρ is the risk-adjusted discount rate, $\mathbb{E}_t$ denotes the expectation operator with information up until time t, $\max(a,\ b)$ denotes the maximum of the quantities a and b, and $\min\{a, b\}$ denotes the minimum of the quantities a and b.

The valuation problem of the capped European call Eq. (1) is also known in the literature as a Bull Call Spread. Note that capping the states of nature where the value of the call exceeds the cap M is equivalent to being long a standard European call option with strike $K^{A,B}$ and short a standard European call option with strike M written on the underlying $S^{A,B}(t)$. Hence

$$C_p^{A,B}\left(S^{A,B}, M, t; T, K^{A,B}\right) = C_p^{A,B}\left(S^{A,B}, \infty, t; T, K^{A,B}\right) - C_p^{A,B}\left(S^{A,B}, \infty, t; T, M\right), \tag{2}$$

where the standard European call $C^{\cdot,\cdot}(S^{\cdot,\cdot}, \infty, t; T, K^{\cdot,\cdot}) = e^{-\rho(T-t)}\times \mathbb{E}_t[max(S^{\cdot,\cdot}(T) - K^{\cdot,\cdot}, 0)]$, i.e. is given by Eq. (1) with $M=\infty$.

Generally, rights to interconnector capacity are sold over a period of time that covers a number of years and represents a significant proportion of the life of the interconnection assets. For expository purposes we will assume that the rights are in the form of a one-year lease and we value a lease for capacity of 1 MWh during peak times and 1 MWh during off-peak times. The value of the interconnector lease is given by the sum of all the capped European call options (one for every day of transmission from A to B and from B to A) between time t and expiry of the lease contract. Denoting by $V(t)$ the value of the interconnector lease with 1 MWh of capacity at time t for 1 h during peak and 1 h during off-peak we have that

$$V(t) = V^{\text{peak}}(t) + V^{\text{off-peak}}(t) \tag{3}$$

[5] See www.nymex.com for more information on the exchange traded spread options in energy commodities.

[6] Bunn and Martoccia (2010) show that such optionality is exhibited by some of the established transmission auction prices for inter-country electricity trading in Europe.

[7] Keppo and Lu (2003) consider the impact that market entry of a large electricity producer has on equilibrium prices.

[8] If the peak time is 12 h then the owner of the interconnector capacity holds 12 call options $C_p^{A,B}$ for the peak time and 12 options for the off-peak time $C_{op}^{A,B}$ where the subscript op stands for off-peak.

where

$$V^{\text{peak}}(t) = \sum_{i=1}^{365} C_p^{\text{A,B}}\left(S^{\text{A,B}}, M, t; t + i/365, K^{\text{A,B}}\right) + \sum_{i=1}^{365} C_p^{\text{B,A}}\left(S^{\text{B,A}}, M, t; t + i/365, K^{\text{B,A}}\right) \tag{4}$$

is the value of the strip of peak real options, and

$$V^{\text{off}-\text{peak}}(t) = \sum_{i=1}^{365} C_{op}^{\text{A,B}}\left(S^{\text{A,B}}, M, t; t + i/365, K^{\text{A,B}}\right) + \sum_{i=1}^{365} C_{op}^{\text{B,A}}\left(S^{\text{B,A}}, M, t; t + i/365, K^{\text{B,A}}\right) \tag{5}$$

is the value of the strip of off-peak real options. Here the notations C_p and C_{op} denote the capped calls on the peak and off-peak segments of every day respectively and the sum is from day 1 until day 365. Hence the one-year lease consists of 1460 options, of which 730 are for a one-hour slot during peak times and 730 are for a one-hour slot during off-peak times.[9] Are the values (4) and (5) arbitrage free? We know that storing electricity in an economical way is not possible, therefore the four strips of 365 options described here cannot be arbitraged using a buy-and-hold argument. Below we show that by setting a simple strategy based on forward contracts, one can derive lower bounds for the four options discussed here.

5. No-Arbitrage bounds

Although the real option valuation of the interconnector requires knowledge of the distribution of the difference between peak and off-peak prices under the statistical measure and the risk-adjusted rate ρ in order to discount the risky cash-flows, one can check whether the strip of call options being used in the valuation satisfies no-arbitrage lower bounds given by the forward or futures markets in both locations.

Assume that the lessor sells capacity for each hour of the day. For example, one can purchase interconnector capacity for the hour 8am to 9am for as many days as desired, or one can purchase the entire peak segment for as many days as desired. Now, let us focus on the no-arbitrage bound satisfied by interconnector capacity on peak electricity. Denote an electricity future for peak electricity in location $i = \{\text{A, B}\}$ by $F_p^i(t, T_1, T_2)$ where t is the current time, T_1 is the expiry of the contract, delivery of electricity starts at time T_1+1, and T_2 is the last day of delivery. Below we show that at time t the price of interconnector capacity between dates T_1+1 and T_2, inclusive, must satisfy

$$\sum_{j=T_1+1}^{T_2} C_p^{\text{A,B}}(t;j) + C_p^{\text{B,A}}(t;j) \geq \sum_{j=T_1+1}^{T_2} e^{-r(j-t)}\left(F_p^{\text{B}}(t,T_1,T_2) - F_p^{\text{A}}(t,T_1,T_2) - K^{\text{A,B}}\right) \tag{6}$$

where $C_p^{\text{A,B}}(t;j) = C_p^{\text{A,B}}(S^{\text{A,B}}, M, t;T, K^{\text{A,B}})$ and $C_p^{\text{B,A}}(t;j) = C_p^{\text{B,A}}(S^{\text{B,A}}, M, t;T, K^{\text{B,A}})$ are the prices of the capped options at time t that give the holder the right, but not the obligation, to use the interconnector to deliver 1 MWh of peak electricity from location A to B, or from B to A, at time T. r is the risk-free rate, and $K^{\text{A,B}}$ and $K^{\text{B,A}}$ are the transmission costs incurred when dispatching the 1 MWh of electricity.

Inequality (Eq. (6)) is a no-arbitrage bound because if it is not satisfied the following set of trades produces a riskless profit. First, assume that market quotes reveal that $F_p^{\text{B}}(t, T_1, T_2) - F_p^{\text{A}}(t, T_1, T_2) - K^{A,B} > 0$; the interconnector capacity between locations A and B for peak electricity costs

$$\sum_{j=T_1+1}^{T_2} C_p^{\text{A,B}}(t;j) + C_p^{\text{B,A}}(t;j); \tag{7}$$

and inequality (Eq. (6)) is not satisfied. Second, pay (Eq. (7)) for the strip of calls on the interconnector capacity for the peak hours between T_1+1 and T_2 and, at the same time, go long a forward contract in location A and short a forward contract in location B (both with expiry T_1 and end of delivery T_2 for peak electricity). Every day, from T_1+1 to T_2, collect the 1 MWh bought at price $F_p^{\text{A}}(t, T_1, T_2)$ in location A, send the power to B via the interconnector, sell it in B for $F_p^{\text{B}}(t, T_1, T_2)$, and pay transmission charges of $K^{\text{A,B}}$. Therefore the present value of the net profits, where we include the cost of the interconnector capacity, is given by

$$\sum_{j=T_1+1}^{T_2} e^{-r(j-t)}\left(F_p^{\text{B}}(t,T_1,T_2) - F_p^{\text{A}}(t,T_1,T_2) - K^{\text{A,B}}\right) - \sum_{j=T_1+1}^{T_2} C_p^{\text{A,B}}(t;j) + C_p^{\text{B,A}}(t;j) > 0, \tag{8}$$

which is greater than zero and represents a riskless profit, i.e. an arbitrage.

One of the points we clarify in the arbitrage strategy above is that although we assume that electricity is bought in market A and sold in market B, the arbitrageur's strategy requires him to purchase options to send peak electricity from both A to B, and also from B to A, even if he never transmits power from B to A. His arbitrage strategy commits him to use all the capacity every day during peak hours in only one direction, and the seller of the capacity charges the amount (Eq. (7)) regardless. This explains why we include the strip of options $\sum_{j=T_1+1}^{T_2} C_p^{\text{B, A}}(t;j)$ as part of the cost of using the interconnector.

Therefore, given two peak forward contracts in locations A and B with the same T_1 and T_2, the following bounds must be obeyed for $t < T_1$:

$$\sum_{j=T_1+1}^{T_2} e^{-r(j-t)}\left(F_p^{\text{B}}(t,T_1,T_2) - F_p^{\text{A}}(t,T_1,T_2) - K^{\text{A,B}}\right) \leq \sum_{j=T_1+1}^{T_2} C_p^{\text{A,B}}(t;j) + C_p^{\text{B,A}}(t;j), \tag{9}$$

and

$$\sum_{j=T_1+1}^{T_2} e^{-r(j-t)}\left(F_p^{\text{A}}(t,T_1,T_2) - F_p^{\text{B}}(t,T_1,T_2) - K^{\text{B,A}}\right) \leq \sum_{j=T_1+1}^{T_2} C_p^{\text{A,B}}(t;j) + C_p^{\text{B,A}}(t;j), \tag{10}$$

for peak hours. Similarly, given off-peak forward contracts in locations A and B, F_{op}^{A} and F_{op}^{B}, we can obtain no-arbitrage lower bounds for the off-peak real options, $C_{op}^{\text{A,B}}(t;j)$ and $C_{op}^{\text{B,A}}(t;j)$.

6. A model for the electricity spot price differentials

Modeling electricity prices, and other financial instruments related to this market, is quite recent in the academic literature. For instance, the work of Schwartz (1997) and Schwartz and Smith (2000) which considered storable commodities served as a platform for a number of articles that proposed no-arbitrage models for the dynamics of electricity prices, see Roncoroni (2010). Examples of no-arbitrage

[9] The value of a one-year lease for the 12 peak and 12 off-peak hourly slots of the day is given by $12(V^{\text{peak}}(t) + V^{\text{off}-\text{peak}}(t))$.

models are found in Cartea and Figueroa (2005) and Lucía and Schwartz (2002). Other models examined in the literature are the so-called equilibrium and hybrid models, see for example: Barlow (2002), Bessembinder and Lemmon (2002), Cartea and Villaplana (2008), and Pirrong and Jermakyan (2008), among others.

Since the valuation of the call options embedded in the interconnector capacity is cast within the real options framework, the emphasis must be placed on a model that is specified under the statistical measure. Instead of estimating the parameters for the two markets A and B, we can value the interconnector capacity by modeling the difference in prices directly. Therefore, we can estimate the parameters of the spread model and use it as the departure point to value the European call options on the spread.[10]

Here we propose a model for the spread in the spirit of the no-arbitrage spot price models which captures the most important features of the price dynamics, that is: large price spikes or jumps, strong mean reversion of large deviations and the presence of a seasonal component. In addition, we obtain the following three desired properties. First, the spread model also exhibits the stylized characteristics observed in the price difference between two locations, specifically large positive and negative deviations that mean revert very quickly to a seasonal trend. Second, the estimation of the spread model parameters can be achieved with the usual techniques. Third, the spread model specification enables us to calculate the price of European-style options by employing standard tools.

Let $S^{A,B}(t) - S^{A}(t) - S^{B}(t)$ denote the spread in wholesale prices at time t between locations A and B. We propose, under the statistical measure, the following arithmetic model for the price differences between locations A and B, $S^{A,B}(t)$, at time T:

$$S^{A,B}(T) = f(T) + X(T) + Y(T) \tag{11}$$

where $f(T)$ is a deterministic seasonal pattern (i.e. the long term trend of the spot) evaluated at time T, $X(T)$ is a mean reverting stochastic process at time T given by

$$X(T) = X(t)e^{-\alpha(T-t)} + \int_t^T e^{-\alpha(T-u)}\sigma(u)dW(u), \tag{12}$$

and $Y(T)$ is a zero-mean reverting pure jump process at time T expressed as

$$Y(T) = Y(t)e^{-\beta(T-t)} + \int_t^T e^{-\beta(T-u)}dJ^{+}(u) + \int_t^T e^{-\beta(T-u)}dJ^{-}(u), \tag{13}$$

where α and β are the speeds of mean reversion for the Gaussian diffusion and the jump process, respectively; $\sigma(t)$ is the time-dependent deterministic volatility; $dW(u)$ are the increment of a standard Brownian motion; and $dJ^{+}(u)$ and $dJ^{-}(u)$ are the increments of a compound Poisson process define as

$$J^{s}(t) = \sum_{n=1}^{N^{s}(t)} j_n^{s}, \quad s = +, -, \tag{14}$$

where $N^{s}(t)$ denotes an inhomogeneous Poisson process with the time-dependent intensity $\lambda^{s}(t)$. The random variables $\{j_1^s, j_2^s, \ldots, j_n^s\}$ represent the size of the jumps in the spread process, which are i.i.d. and exponentially distributed with parameter η^{s} so that the expected size of the jumps is $1/\eta^{s}$.

The spread between B and A is given by $S^{B,A}(T) = -S^{A,B}(T)$. Once we have the model for the price differences, we can proceed to value the European call options on the spread.

[10] Models for the spread can also be found in Benth and Saltyte-Benth (2006) and Benth and Kufakunesu (2009).

6.1. Call option with jumps

In this subsection we describe how to price the real option (Eq. (1)) when the spread follows (Eq. (11)), with OU component (Eq. (12)), and jump component (Eq. (13)) with exponentially distributed jumps. The value of the call option is expressed in closed-form in Fourier space (see Appendix for details). The value of a European-style option to transmit electricity from market B to market A is given by evaluating

$$C^{A,B}\left(S^{A,B}, t; T, K^{A,B}\right) = \frac{e^{-r(T-t)}}{2\pi}\int_{-\infty+i\xi_i}^{\infty+i\xi_i}\Psi_S^{A,B}(-\xi)\Pi^{A,B}(\xi)d\xi \tag{15}$$

where the transform variable $\xi = \xi_r + i\xi_i$, with $\xi_r, \xi_i \in \mathbb{R}$, $i = \sqrt{-1}$, and $\Pi^{A,B}(\xi)$ is the Fourier transform of the call option payoff between locations A and B:

$$\Pi^{A,B}(\xi) = \int_{-\infty}^{\infty} e^{i\xi x} max\left(x - K^{A,B}, 0\right)dx = -\frac{e^{i\xi K^{A,B}}}{\xi^2}, \quad \text{for} \quad \xi_i > 0. \tag{16}$$

To calculate the inversion in Eq. (15) we also require the characteristic function of $S^{A,B}(T)$ (see Appendix for the proof):

$$\Psi_S^{A,B}(\xi) = e^{i\xi h(T) - \frac{1}{2}\xi^2\int_t^T e^{-2\alpha(T-u)}\sigma^2(u)du} \times e^{\int_t^T\left(\frac{\eta_1}{\eta_1 - i\xi e^{-\beta(T-u)}} - 1\right)\lambda^{+}(u)du} \times e^{\int_t^T\left(\frac{\eta_2}{\eta_2 + i\xi e^{-\beta(T-u)}} - 1\right)\lambda^{-}(u)du}, \tag{17}$$

where $h(T) = f(T) + X(t)e^{-\alpha(T-t)} + Y(t)e^{-\beta(T-t)}$, $\lambda^{+}(t)$ and $\lambda^{-}(t)$ are the time-dependent intensities of the Poisson arrival of positive and negative jumps respectively, and we require $-\eta_2 < \xi_i < \eta_1$.

Note that if we make the assumption that there are no jumps in the spread, the value of the capped European option to transmit electricity from location B to A is given by (see Appendix for the proof):

$$C^{A,B} = e^{-\rho(T-t)}\left\{\left[\mu(t,T) - K^{A,B} + \frac{v(t,T)\phi(\beta_1)}{1-\Phi(\beta_1)}\right]\Phi(-\beta_1) - \left[\mu(t,T) - M + \frac{v(t,T)\phi(\beta_2)}{1-\Phi(\beta_2)}\right]\Phi(-\beta_2)\right\}, \tag{18}$$

where

$$\mu(t,T) = f(T) + X(t)e^{-\alpha(T-t)} \quad \text{and} \quad v^2(t,T) = \int_t^T e^{-2\alpha(T-u)}(\sigma(u))^2du, \tag{19}$$

$C^{A,B} = C^{A,B}(S^{A,B}, M, t; T, K^{A,B})$, $\phi(x)$ and $\Phi(x)$ denote the probability density and distribution functions of a standard normal random variable. $\beta_1 = (K^{A,B} - \mu(t,T))/v(t,T)$, $\beta_2 = (M - \mu(t,T))/v(t,T)$ and M is the liquidity cap. The price of the option to transmit electricity from A to B is calculated in the same way.

7. Estimation of model parameters

In this section, we discuss how we estimate the underlying structural parameters of the state variables $X(t)$ and $Y(t)$, as well as the deterministic seasonal factor, $f(t)$. The estimation procedure requires the following steps. First, find the deterministic seasonal trend $f(t)$ using an OLS regression and compute the detrended spread. Second, detect the positive and negative jumps in the detrended spread series considering mean reversion in the jumps. The jump detection algorithm we employ is designed to cope with the problem of miss-identifying mean reversion as jumps. Third, find the MLE of the (possibly time-dependent) intensity of positive and negative jumps. Finally, estimate the parameters of the state variables $X(t)$ and $Y(t)$.

Eqs. (11)–(13) describe the continuous-time model for the spread $S^{A,B}(t)$ between locations A and B. We estimate the parameters of the discrete-time versions of the continuous-time Eqs. (12) and (13) employing daily electricity data. For ease of notation we use time t in subscript to denote the discreteversion of the continuous time variables, for example the discrete versions of $X(t)$ and $Y(t)$ are denoted by X_t and Y_t respectively. To estimate the parameters of the Gaussian process $X(t)$ we use the discrete model

$$X_t = e^{-\alpha\Delta t}X_{t-\Delta t} + \varepsilon_t \qquad t = 1, 2, \ldots, N \tag{20}$$

where ε_t satisfies

$$\mathbb{E}\left[\varepsilon_t^2\right] = \sigma_{\varepsilon,t}^2 = \frac{\sigma_t^2}{2\alpha}\left(1-e^{-2\alpha\Delta t}\right) \quad \text{and} \quad \mathbb{E}[\varepsilon_t] = 0, \tag{21}$$

and we specify the discrete-time version of the jump factor $Y(t)$ as

$$Y_t = e^{-\beta\Delta t}Y_{t-\Delta t} + \Delta J^+ + \Delta J^-, \tag{22}$$

where $\Delta J^{+,-} = J^{+,-}\Delta N^{+,-}$. Here ΔN^+ and ΔN^+ are the increments of the discrete-time counting process for positive and negative jumps with arrival frequency $\lambda_t^+\Delta t$ and $\lambda_t^-\Delta t$. The random variables J^+ and J^- are positive and negative exponentially i.i.d. jump sizes with parameters η_+ and η_-. Hence, their means are $\mathbb{E}[J^+] = 1/\eta_+$ and $\mathbb{E}[J^-] = 1/\eta_-$.

We use these discrete schemes to estimate the set of parameters of the Gaussian, $\{\alpha, \sigma(t)\}$, and jump processes, $\{\beta, \lambda_t^+(t), \lambda_t^-(t), \eta^+, \eta^-\}$. Before determining these parameters, we proceed with the analysis of the seasonal behavior of spreads and prices.

7.1. The deterministic function $f(t)$ and other seasonal features

We follow the approach developed in Jaillet et al. (2004) and Manoliu and Tompaidis (2002) to model the seasonal component $f(t)$ of the spread of electricity spot prices. The discrete-time version of $f(t)$ takes the form:

$$f_t = f_0 + f_1 t + \sum_{m=1}^{11} F_m D_t^m \tag{23}$$

where f_0 is a constant, f_1 is the coefficient for the time trend, F_m, for $m = 1, \ldots, 11$, are constant parameters and D_t^m are monthly dummies taking the value 1 if t belongs to the m-th month and 0 otherwise. Table 2 shows the estimates and Fig. 1 depicts some examples of the seasonal function (23) for individual peak and off-peak spreads.

One of the features that we explore is whether there is a seasonal pattern in the volatility of the spread innovations which can emerge from a seasonal pattern in the diffusion coefficient $\sigma_{\varepsilon,t}$ of the X_t process; and/or in the intensity of the jumps, λ_t^+ and λ_t^-. To test for seasonality in the parameters that drive the conditional variance, we assume a time-dependent seasonal functional form for the volatility $\sigma_{\varepsilon,t}$, and for the intensity parameters λ_t^+ and λ_t^-.

In order for the number of parameters in the model to be tractable, and for us to have sufficient observations for each case, we model the periodicity with seasonal dummies as follows:

$$\sigma_{\varepsilon,t} = \sigma_{\varepsilon,\text{winter}} \cdot D_t^{\text{winter}} + \sigma_{\varepsilon,\text{spring}} \cdot D_t^{\text{spring}} + \sigma_{\varepsilon,\text{summer}} \cdot D_t^{\text{summer}} + \sigma_{\varepsilon,\text{autumn}} \cdot D_t^{\text{autumn}} \tag{24}$$

Table 2
Coefficients of the Deterministic Component. In this table we report the OLS estimate parameters of the time-dependent deterministic component of Eq. (23): $f_t = f_0 + f_1 t + \sum_{m=1}^{11} F_m D_t^m$, where f_0 is a constant, f_1 is the coefficient for the time trend, F_m for $m = 1, \ldots, 11$ are constant parameters, and D_t^m are monthly dummies, taking the value 1 if t belongs to the mth month and 0 otherwise. R^2 shows the goodness of fit of the linear regression, the F-statistic and its p-value, in parentheses, show the overall significance of the model. DW stands for the Durbin–Watson statistic (p-value in parentheses), and is used to test the presence of autocorrelation in the residuals.

	France–Germany				France–UK				Nord Pool–Germany			
	Peak		Off-peak		Peak		Off-peak		Peak		Off-peak	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat
f_0	2.24	1.51	−5.94	−3.77	2.67	1.17	0.56	0.42	3.36	1.67	4.51	2.65
f_1 ($\times 10^2$)	0.19	3.81	0.35	6.56	−0.23	−3.03	−0.51	−11.89	−0.89	−16.18	−0.25	−5.47
Jan	−3.68	−1.95	2.12	1.06	4.07	1.47	0.79	0.50	−1.22	−0.48	0.64	0.30
Feb	−3.63	−1.88	0.82	0.40	2.37	0.84	1.60	0.98	−0.15	−0.06	−3.62	−1.64
Mar	−3.70	−1.95	0.70	0.35	−2.51	−0.91	−1.32	−0.83	1.33	0.52	−2.94	−1.37
Apr	−5.98	−3.14	−0.38	−0.19	1.68	0.60	−2.69	−1.67	3.04	1.19	−0.69	−0.32
May	−6.05	−3.20	−1.14	−0.57	−0.08	−0.03	−9.21	−5.59	2.71	1.07	−1.11	−0.52
Jun	−6.45	−3.38	−3.61	−1.79	9.00	3.11	−9.54	−5.75	−3.32	−1.31	−2.29	−1.07
Jul	−4.24	−2.17	0.49	0.24	14.26	4.98	−12.02	−7.30	−13.04	−5.13	−3.48	−1.62
Aug	−8.10	−4.14	−4.74	−2.29	1.02	0.36	−9.86	−5.99	2.83	1.11	1.27	0.59
Sep	−5.89	−2.98	−2.03	−0.97	3.36	1.16	−2.84	−1.71	−2.62	−1.02	−1.72	−0.79
Oct	−4.93	−2.53	−0.89	−0.43	1.18	0.41	1.10	0.67	−4.31	−1.70	−1.52	−0.71
Nov	−3.03	−1.55	1.36	0.66	5.23	1.81	1.11	0.67	−10.50	−4.09	−0.39	−0.18
R^2	0.019		0.031		0.034		0.113		0.126		0.018	
f-stat	3.207		5.192		5.538		28.262		28.110		3.543	
	(0.000)		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)	
DW	2.167		2.197		1.394		1.105		1.028		1.407	
	(0.001)		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)	

	Germany–The Netherlands				The Netherlands–UK			
	Peak		Off-peak		Peak		Off-peak	
	Coeff	t-stat	Coeff	t-stat	tCoeff	t-stat	Coeff	t-stat
f_0	−30.12	−9.35	7.68	5.07	24.05	6.82	−7.51	−4.99
f_1 ($\times 10^2$)	0.76	8.71	0.07	1.82	−1.17	−10.24	−0.65	−13.42
Jan	12.03	2.97	−4.85	−2.54	−1.72	−0.40	5.26	2.89
Feb	14.29	3.45	−6.41	−3.29	−5.25	−1.21	8.80	4.75
Mar	12.69	3.11	−6.45	−3.37	−8.74	−2.04	5.71	3.13
Apr	12.65	3.07	−6.13	−3.17	−1.63	−0.38	5.72	3.09
May	9.94	2.44	−3.71	−1.94	−1.86	−0.42	−3.81	−2.03
Jun	6.18	1.53	−3.62	−1.91	11.56	2.60	−4.97	−2.63
Jul	8.70	2.15	−5.55	−2.91	4.33	0.98	−10.62	−5.66
Aug	3.98	0.98	−5.47	−2.88	9.25	2.11	−3.79	−2.02
Sep	1.25	0.30	−2.91	−1.51	3.30	0.75	0.25	0.13
Oct	2.51	0.62	−2.52	−1.33	10.10	2.31	2.87	1.54
Nov	3.43	0.84	−2.22	−1.16	9.97	2.25	−0.25	−0.13
R^2	0.048		0.012		0.078		0.168	
f-stat	9.630		2.247		13.115		31.486	
	(0.000)		(0.000)		(0.000)		(0.000)	
DW	1.420		1.885		1.173		0.563	
	(0.000)		(0.002)		(0.000)		(0.000)	

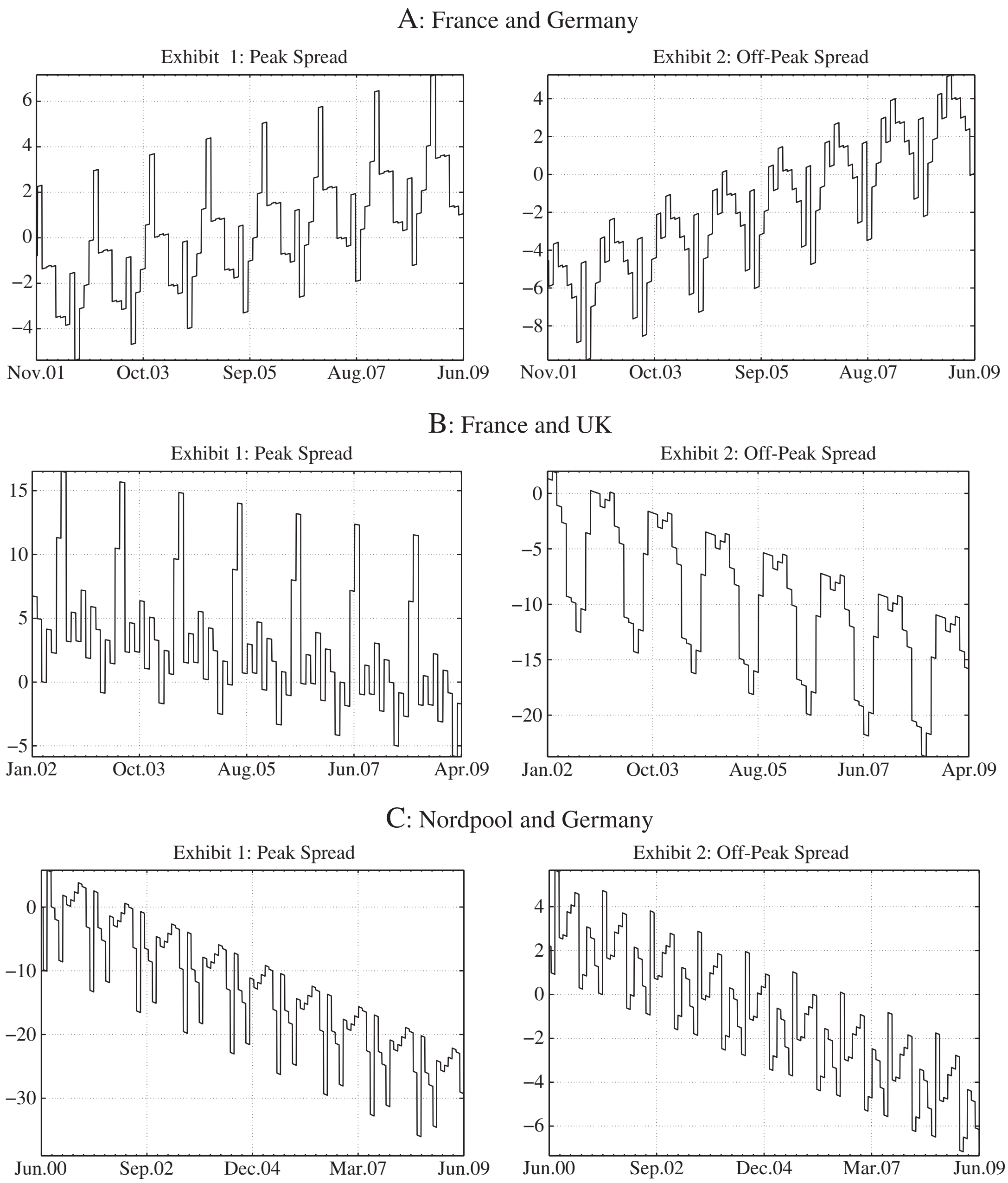

Fig. 1. Examples of Interconnection's Deterministic Component for Peak and Off-Peak Load. This figure shows the deterministic component of the peak and off-peak spread $S^{i,j}$ between the respective locations (black solid line in Exhibits 1 and 2 of each Panel). Table 2 reports the parameters employed to depict the seasonal components. In all graphs the y-axis is in Euros/MWh.

for the volatility, and

$$\lambda_t^{+,-} = \lambda_{\text{winter}}^{+,-} \cdot D_t^{\text{winter}} + \lambda_{\text{spring}}^{+,-} \cdot D_t^{\text{spring}} + \lambda_{\text{summer}}^{+,-} \cdot D_t^{\text{summer}} + \lambda_{\text{autumn}}^{+,-} \cdot D_t^{\text{autumn}} \tag{25}$$

for the intensity parameters of the positive and negative jumps. $D_t^{\cdot}$ are seasonal dummies which take the values 0 or 1. That is, December, January, and February correspond to the dummy variable D_t^{winter}; March, April, and May to D_t^{spring}; June, July, and August to D_t^{summer}; and, finally, September, October, and November to D_t^{spring}.

In the next section we show how to deal with these seasonal patterns in trend, volatility and intensity and their relation with the filtering of jumps and the estimation of the state-variables.

7.2. Methodology to deal with the jump process and parameter estimates

Here we discuss how to estimate the underlying structural parameters of the Gaussian variable $X(t)$ and the jump factor $Y(t)$ described by the discrete-time models (20) and (22). The estimation consists of the next steps:

i) Remove the fitted seasonal component from the spread and calculate the detrended spread denoted by $\tilde{S}_t^{A,B} = S_t^{A,B} - f_t$.
ii) Detect the arrival of jumps (positive or negative) in the detrended spread series $\tilde{S}_t^{A,B}$ and avoid classifying fast mean-reversion as jumps.
iii) Estimate the jump intensities allowing for seasonal dependence in the arrival of the jumps.

Table 3
Intensity parameter of the jumps. This table reports the intensities of jump arrivals λ. We model positive and negative jumps assuming two cases, one where the two intensities are constant throughout the year, and the other with time-dependent intensities λ(t) which vary with the season of the year, i.e. $\lambda_t^{+,-} = \lambda_{\text{winter}}^{+,-} \cdot D_t^{\text{winter}} + \lambda_{\text{spring}}^{+,-} \cdot D_t^{\text{spring}} + \lambda_{\text{summer}}^{+,-} \cdot D_t^{\text{summer}} + \lambda_{\text{autumn}}^{+,-} \cdot D_t^{\text{autumn}}$, where D_t^{winter}, D_t^{spring}, D_t^{summer} and D_t^{autumn} are seasonal dummies. *p-val (+,−)* are the *p*-values of likelihood ratio tests, where the unrestricted model is the one with seasonal lambda. *p*-values under 0.050 indicate rejection of the constant (restricted) model at a 5% level of significance. For example, we see that for peak spread between France and Germany, we reject the constant model in favor of a seasonal intensity for both positive (5% significance) and negative jumps (10% significance).

	France–Germany				France–UK				Nord Pool–Germany				Germany–The Netherlands				The Netherlands–UK			
	Peak		Off-peak		Peak		Off-peak		Peak		Off-peak		Peak		Off-peak		Peak		Off-peak	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	tCoeff	t-stat	Coeff	t-stat
λ^+	5.976	5.943	5.976	5.943	7.520	6.630	5.560	6.760	2.458	3.916	4.382	5.496	7.030	7.138	6.706	6.951	8.431	7.014	2.007	3.018
λ^-	5.849	5.871	6.230	6.086	5.673	5.648	4.241	5.797	8.015	7.731	5.237	6.086	7.138	7.199	3.461	4.761	6.558	6.086	8.833	7.199
λ^+_{winter}	7.814	3.141	6.837	2.892	7.127	2.892	4.741	2.760	3.910	2.182	4.345	2.336	6.655	3.018	8.873	3.600	6.783	2.760	1.043	0.750
λ^+_{spring}	1.924	1.256	4.328	2.182	2.515	1.470	3.231	2.182	0.854	0.750	4.698	2.483	2.184	1.470	4.804	2.483	3.098	1.666	0.516	0.429
λ^+_{summer}	7.842	3.018	8.888	3.261	9.882	3.377	7.435	3.490	1.254	1.020	6.687	3.141	7.140	3.261	7.980	3.490	9.374	3.261	2.757	1.470
λ^+_{autumn}	6.574	2.624	3.835	1.848	11.077	3.600	7.121	3.377	3.877	2.182	1.723	1.256	12.062	4.402	5.169	2.624	14.954	4.308	3.877	1.848
λ^-_{winter}	5.372	2.483	9.279	3.490	6.109	2.624	2.188	1.666	8.690	3.600	6.952	3.141	4.880	2.483	3.106	1.848	7.304	2.892	5.217	2.336
λ^-_{spring}	3.847	2.020	3.366	1.848	3.018	1.666	1.436	1.256	5.125	2.624	3.417	2.020	1.747	1.256	3.057	1.848	3.098	1.666	5.680	2.483
λ^-_{summer}	9.934	3.490	8.888	3.261	3.294	1.666	8.217	3.708	8.358	3.600	7.522	3.377	7.980	3.490	6.300	3.018	3.860	1.848	15.440	4.402
λ^-_{autumn}	4.383	2.020	3.287	1.666	10.523	3.490	5.538	2.892	9.908	3.916	3.015	1.848	13.785	4.761	1.292	1.020	12.185	3.813	9.415	3.261
p-val (+)	0.019		0.154		0.005		0.110		0.040		0.060		0.001		0.237		0.001		0.070	
p-val (−)	0.074		0.012		0.011		0.000		0.270		0.056		0.000		0.032		0.003		0.004	

iv) Estimate the parameters of the exponential distributions for positive and negative jumps.
v) Estimate the mean-reversion rates of the Gaussian and jump factors of the spread model and the volatility of the Gaussian process.

The first step is straightforward, we discuss the others below.

7.2.1. Detecting jumps

We apply a recursive semi-parametric filter to identify the calendar position of the jumps in the spread. The procedure identifies a hypothetical arrival of a jump when the detrended spread difference deviates, in absolute value, by more than three standard deviations from its mean. In our framework, these standard deviations might be time-dependent, so we compute them taking into account the possible seasonal pattern in the variance of the spread, see Eq. (24). We remove the observations identified as possible jumps, recalculate the mean, and proceed to filter again. We iterate until no jumps are found. However, in highly mean-reverting time series, for instance electricity prices, some of these hypothetical jumps could correspond to the mean-reversion effect and not to the arrival of jumps of opposite sign. As we can see in Eq. (22), the mean-reversion effect with rate β is always present in the discrete-time dynamics of the jump process Y_t.

7.2.2. Disentangling jumps from mean reversion

Our methodology is designed to cope with the problem of miss-identifying mean reversion as jumps as follows. Assume that at time t spread prices are above the estimated seasonal trend $S_t^{A,B} > \hat{f}_t$. Suppose that the next price innovation is negative, i.e. $S_{t+1}^{A,\,B} - S_t^{A,\,B} < 0$, and that it is flagged as a possible jump because in absolute terms the innovation is larger than three standard deviations from the mean. Now, before labeling it as a downward jump, we check "how close" the new value $S_{t+1}^{A,\,B}$ is to $\hat{f}_{t+1}$. If it is close to the seasonal component then it cannot be a downward jump, it must be mean reversion. If it is sufficiently below $\hat{f}_{t+1}$, then it is a downward jump, that is, it diverges from the long-term trend. The metric used here to assess proximity to the seasonal trend is based on the confidence bounds for the estimates of $\hat{f}_t$. Spread prices that after a large negative innovation fall within 95% of the confidence bounds of the estimated seasonal trend, $\hat{f}_t$, or above are not considered negative jumps, but are due to mean reversion. Only innovations that take the spread below the lower 95% confidence interval are considered a downward jump. Similarly, large upward shocks to the spread are considered positive jumps if they satisfy two requirements: the innovation is larger than three standard deviations from the mean; and the resulting spread at time $t+1$ is above the upper 95% confidence interval of the estimated seasonal trend, $\hat{f}_t$.

With the advent of high-frequency financial data, jump detection in equity returns has received a considerable amount of attention. Perhaps the first non-parametric filter designed to detect the arrival (including position) of Poisson jumps is that of Lee and Mykland (2008). In order to appreciate the performance of our jump filter we use Monte Carlo simulations and compare our results to those given by the Lee–Mykland ($L-M$) test. For example, we simulate 500 paths of the discrete version of Eq. (11) where we use the seasonal component estimated for the peak spread between France and Germany. [11]

We compare the performance of the two tests by looking at the probability that the test fails to detect a jump, and the probability of spurious detection. We find that our jump detection filter performs better than the L–M test on both counts. In this example, the probability of not detecting an actual jump (resp. detecting a spurious jump) with our filter is 0.17 (resp. 0.03), whereas for the L–M is 0.48 (resp. 0.44).[12]

7.2.3. Estimating jump intensities

Once we have identified the jumps, we estimate their intensity by maximum likelihood. We analyze two scenarios, one where intensities are constant, and the other where intensities are time-dependent and may exhibit a seasonal pattern as described in Eq. (25). MLE results for the positive and negative intensities $\lambda^+(t)$ and $\lambda^-(t)$ are shown in Table 3. In the table we also report the *p*-values of the likelihood ratio test, which compares the unrestricted model for the jump intensities (seasonal $\lambda^{+,\,-}(t)$) with the restricted model of constant intensities. *p*-values under 0.10 indicate rejection of the constant (restricted) model with a 10% significance level. In 16 out of 20 cases, we reject the model with constant intensity at a 10% significance level. For example, for peak spread between France and Germany, we reject the constant model in favor of a seasonal intensity for both positive (5% significance) and negative jumps (10% significance). Results in Table 3 show that the estimated number of jumps for spreads ranges from 2 to 9 per year. In

[11] The parameters we employ are: $\alpha=250$, $\beta=150$, λ's are seasonal, being $\lambda^+=[8,2,8,7]$ and $\lambda^-=[5,\ 4,\ 10,\ 4]$ (that is, one $\lambda^{\pm}$ for each season: winter, spring, summer, and autumn), $\eta^+=1/70$, $\eta^-=1/50$, $\sigma=10$, and the length of the time series is $N=2774$. We chose these parameters based on the estimation results reported below.

[12] We note that the L–M test is designed to perform better the higher the frequency of the data and in this case we have used daily observations which is the frequency available for the our electricity data. Moreover, other filters have been proposed in the literature, however, although they are designed to tell whether there is a non-Gaussian component in the price process, they are not capable of detecting the position of the possible jump.

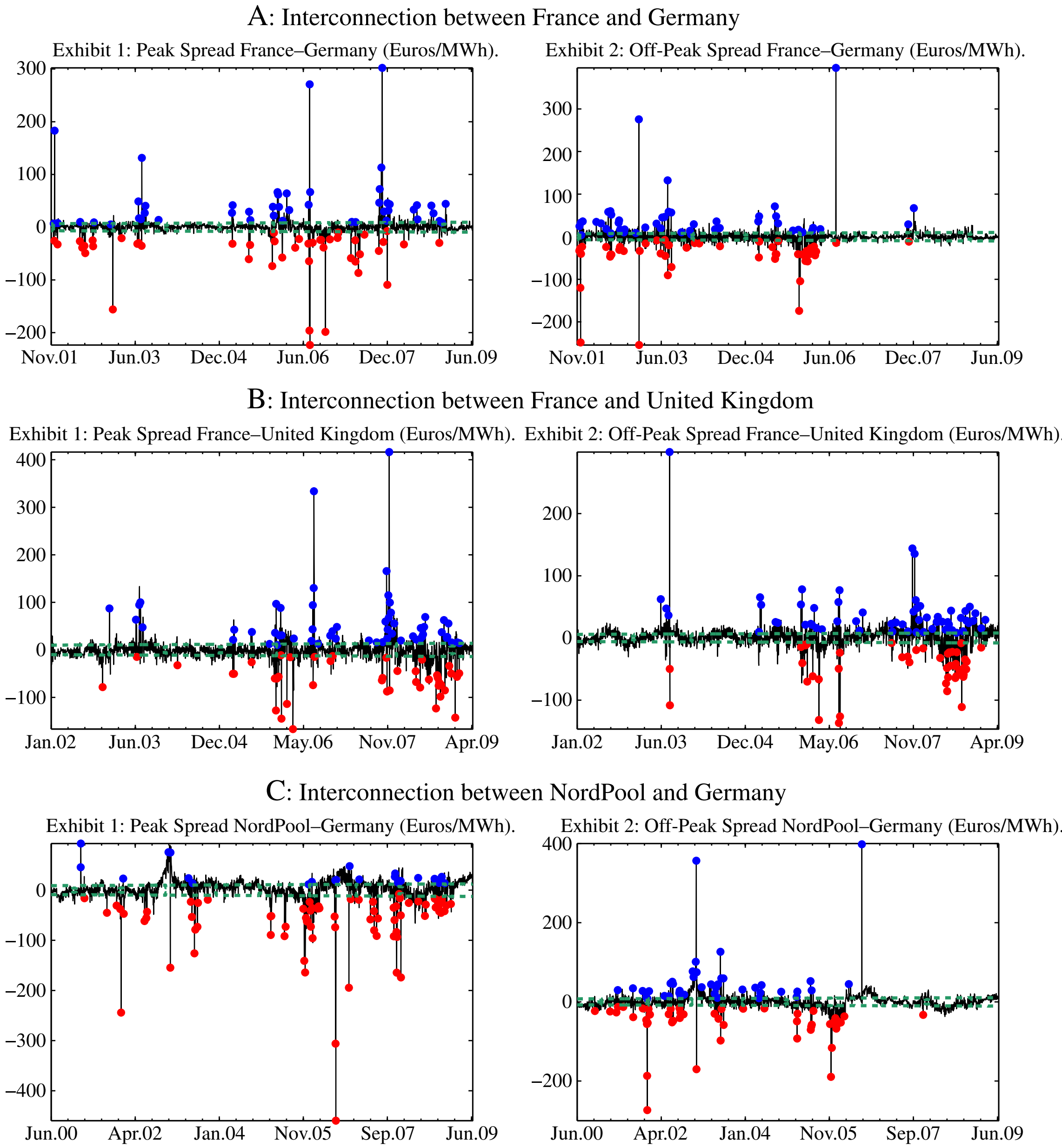

Fig. 2. Examples of Detecting Jumps from the Jump Diffusion Process. This figure shows the spread series $S^{i,j}$ for the two possible load regimes: peak and off-peak (Exhibits 1 and 2, respectively) for three examples of interconnection. The dashed line represents the 95% confidence intervals for the deterministic component for each series. The red circles below the lower confidence bound mark the presence of a negative jump, while the blue circles above the upper confidence bound show the position of a positive jump.

Fig. 2, we show the positions of negative and positive jumps for some of the markets we study.

7.2.4. Estimating jump size parameters

One of the advantages of being able to locate jumps is that we also know their sizes and signages. We use this information to fit an exponential distribution to the jump data and to obtain the average sizes of positive and negative jumps given by $1/\eta^+$ and $1/\eta^-$.[13] The estimates are shown in Table 4.

[13] We do this in the following way. Once we have located the positive and negative jumps, we estimate the sizes using the differences between the spread $\tilde{S}_t^{A,B}$ of the observation and the previous value of the spread, $\tilde{S}_{t-1}^{A,B}$. This approach performs well when the mean-reversion rates of jumps and OU processes are large, which is the case in electricity prices and the spread.

7.2.5. Estimating mean-reversion parameters and the volatility of the OU process

For each time series we estimate the mean-reversion rates of the Gaussian process X_t and jump process Y_t, and the volatility parameters of X_t by minimizing the mean-squared errors which are given by the average of the squared differences between the observed and the modeled spreads. Using the model specification of Eqs. (11)–(13), we have that the differences between observed spreads and the proposed jump dynamics follows an autoregressive process of order one, that is, $\tilde{S}_t^{A,B} - e^{-\beta} Y_{t-1} - \hat{J}_t^+ - \hat{J}_t^- = X_t$, where $X_t = e^{-\alpha} X_{t-1} + \varepsilon_t$ and $\varepsilon_t \sim N(0, \sigma^2_{\varepsilon,t})$. Hence, the estimates of both mean-reversion rates β and α, as well as the volatility of the OU process can be simultaneously obtained by means of nonlinear least squares. We report the results in Table 4 where we see that spreads show significant mean reversion in jumps and in the Gaussian deviations. The half-life of the jumps ranges between 1 and 15 days approximately.

Table 4
Estimates of seasonal volatility, mean reversion rates and jumps sizes. This table reports the parameter estimates of seasonal volatility $\sigma(t)$ of the diffusion process, the jumps sizes mean $1/\eta^{+}$ and $1/\eta^{-}$, and the mean reversion rates α and β of the diffusion and the jump component. All parameters are expressed in annual terms and in Euros/MWh, to be consistent with the arithmetic model for the spread. We show t-statistics and the RMSE of the model.

	France–Germany				France–UK				Nord Pool–Germany				Germany–The Netherlands				The Netherlands–UK			
	Peak		Off-peak		Peak		Off-peak		Peak		Off-peak		Peak		Off-peak		Peak		Off-peak	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	tCoeff	t-stat	Coeff	t-stat
α	273.07	14.02	108.02	14.19	122.05	15.41	109.06	18.58	66.54	16.23	70.21	16.38	5.37	7.57	1.42	4.06	194.44	15.75	50.45	13.95
σ_{winter}	230.17	19.36	300.22	20.89	341.70	21.48	162.47	27.36	288.27	25.63	237.22	25.38	353.00	30.30	275.56	24.67	331.83	20.61	117.53	24.38
σ_{spring}	132.57	19.48	188.81	21.24	245.19	21.59	165.48	27.47	179.57	25.77	181.76	25.70	308.88	30.50	186.01	25.25	277.81	20.72	128.01	24.51
σ_{summer}	289.95	18.84	244.60	20.89	321.42	20.95	239.16	26.67	241.30	25.96	290.71	25.87	785.73	31.02	205.20	25.58	620.57	20.15	180.54	23.83
σ_{autumn}	218.22	18.68	211.21	20.48	370.60	20.89	193.47	26.53	257.66	25.71	190.03	25.63	444.71	30.65	204.41	25.29	458.49	20.26	137.28	23.86
β	183.90	30.12	96.46	21.10	167.14	21.68	121.68	21.38	206.85	25.11	100.45	20.67	10.34	21.43	52.65	15.22	231.53	22.84	120.37	14.12
$1/\eta^{+}$	72.72	5.43	67.77	5.43	86.35	6.12	53.12	6.25	56.14	3.39	74.68	4.98	106.25	6.63	71.45	6.44	100.57	6.50	28.44	2.49
$1/\eta^{-}$	59.24	5.36	80.87	5.57	76.82	5.13	53.27	5.28	71.48	7.22	72.44	5.57	135.45	6.69	79.70	4.24	74.19	5.57	33.35	6.69
RMSE	9.10		10.94		14.89		9.93		12.91		11.92		30.35		12.42		20.25		7.86	

As a specific case we study a “naive” or misspecified version of the spread model where we do not include the jump process $Y(t)$. In the interest of space, the estimates of the model without jumps are not reported, however, results show that the introduction of jumps in the model delivers gains in the in-sample performance of between 20% and 48% compared to the “naive” version.

8. The market value of interconnectors

In this section we discuss the results of valuing interconnection capacity in neighboring European countries. Based on the market data described above, we calculate the market value of a one-year lease of an interconnector that gives the lessee the right, but not the obligation, to transmit 1 MWh of electricity between two markets during peak and off-peak times. The lease contract starts on January 1, 2010 and ends on December 31, 2010 and we assume that the initial condition of the OU and jump processes are both zero: $X(t)=Y(t)=0$ and $t=$ January 1, 2010. We estimate the spread model with electricity prices data available until July 2009. Using these estimates, we value the interconnector lease that starts on January 1, 2010. Since the estimation is done in July 2009, the values for $X(t)$ and $Y(t)$ in January 1, 2010 are unknown. Hence, we take as initial values of the OU and jump processes the expected value of both variables, which is zero for both. We use the valuation formulae (4) and (5) to calculate prices for the four strips of options that total the value of the one-year lease of 1 MWh capacity for a one-hour slot during peak time and a one-hour slot during off-peak time. We assume that the cash-flows are discounted at the risk-adjusted rate $\rho=10\%$.

Initially we do not check whether the option values that we obtain are equal or above the no-arbitrage bounds derived in Section 5. We do this below in Subsection 8.1, where we calculate the lower bounds using futures data. Our assumption is that the lessor will price the real options and calculate the lower bounds (9) and (10) for peak-hours, and the corresponding bounds for off-peak hours, but for clarity we show the results of the option valuation first and then where necessary we amend the values in the light of the no-arbitrage bounds.[14]

We provide different values of the interconnector, which result from different assumptions about: the seasonal function of the spread; the liquidity and depth in both markets; and how jumps affect the extrinsic value of the real options used to calculate the value of the interconnector. We discuss how these three assumptions affect the value of the interconnector.

8.1. Seasonal component

One of the points that we scrutinize is how the seasonal component affects the value of the interconnector. In this paper we have assumed that the seasonal component is deterministic and that the historical seasonal trend will repeat itself at future dates when forecasting the spread. Although this assumption is consistent with that of the literature, care must be taken when producing forecasts and pricing options written on electricity price spreads. Indeed, inspecting Fig. 1 prompts the question of whether it is plausible to expect that the seasonal component will in the future be broadly the same as it was in the past. The answer is probably not, but until now there has been no better alternative in the literature. Therefore, to appreciate the contribution of the seasonal component to the value of the interconnector, we look at different scenarios where we assume that for future dates the seasonal component is as in Fig. 1, and for comparative purposes we also assume that the seasonal component is $f(T)=0$ for all T. To assume that the seasonal trend is zero is an extreme case because the markets that we are considering have their idiosyncrasies and this makes it reasonable to expect predictable patterns in the average price differentials between them.

8.2. Liquidity constraints

As discussed above, it does not seem plausible to exploit large price differentials due to liquidity reasons in the two markets. We cap the maximum price differentials that can be profited from at different levels: $M\in\{10, 20, 30, 40, 50, \infty\}$ Euros/MWh, where we allow $M=\infty$ to include the hypothetical case where there are no liquidity constraints in the day-ahead market. In all examples we assume that the transmission costs from A to B, and from B to A, for both peak and off-peak times, are $K=5$ Euros/MWh, which seems a plausible figure for the interconnection costs according to some market participants. Nevertheless, these costs could vary across different markets and the value of the interconnector will be affected by changes in K: the higher is K the lower the value of the interconnector.

8.3. Jumps and extrinsic value

Both the volatility of the OU component of the spread model and the jumps increase the value of the interconnector. We can isolate the contribution of the jumps to the value of the interconnector in the following way. Estimate the model parameters and value the interconnector with the jumps in the spread and then compare it to the valuation obtained if we “switch off” the jump factor, by setting the jump intensities of the positive and negative jumps to zero.

We address the three points discussed above and report the values of the interconnector in Tables 5 and 6. In the tables the values of the one year-lease are broken into the four options available to the manager of the lease: transmit electricity from A to B and from B to A

[14] If one or more lower bounds are not satisfied, the lessor will set the price according to the bound instead of the price indicated by the value of the strip of options. Alternatively, if a bound is not satisfied, one expects that market participants will bid the price of the lease up until the price reaches the no-arbitrage bound.

Table 5
Value of one-year Interconnector lease. The model for the spread is Eqs. (11), (12), and (13). The parameters of the seasonal component $f(T)$ are given in Table 2; the parameter estimates of the OU, $X(T)$, are in Panel B of Table 4; the parameters of the jump component, $Y(T)$, are at the bottom of Panel B in Table 4 and in Panel B of Table 3. In columns 3 and 4 the intensity parameters, $\lambda^{\pm}_{season}$, of the positive and negative jumps are set to zero.

$S(T)=$	Seasonality, OU, and jumps		Seasonality and OU		Seasonality, OU, and jumps		Seasonality and OU	
	$f(T)+X(T)+Y(T)$		$f(T)+X(T)$		$f(T)+X(T)+Y(T)$		$f(T)+X(T)$	
M	Peak France and Germany				Off-peak France and Germany			
	Ger → Fr	Fr → Ger	Ger → Fr	Fr → Ger	Ger → Fr	Fr → Ger	Ger → Fr	Fr → Ger
10	601	251	572	209	737	459	728	397
20	1136	467	1002	337	1697	1023	1605	805
30	1312	554	1074	355	2177	1316	1960	939
40	1398	605	1081	357	2416	1488	2077	975
50	1457	641	1082	357	2550	1604	2110	984
∞	1681	761	1082	357	2949	2095	2120	985
	Peak France and UK				Off-peak France and UK			
	UK → Fr	Fr → UK	UK → Fr	Fr → UK	UK → Fr	Fr → UK	UK → Fr	Fr → UK
10	616	661	580	662	89	1333	41	1382
20	1472	1568	1339	1540	187	3354	62	3461
30	1960	2058	1717	1975	243	4472	64	4565
40	2229	2304	1880	2159	282	4969	64	5004
50	2382	2430	1940	2,2256	310	5174	64	5143
∞	2850	2725	1967	2255	401	5441	64	5189
	Peak Nord Pool and Germany				Off-Peak Nord Pool and Germany			
	Ger → NP	NP → Ger	Ger → NP	NP → Ger	Ger → NP	NP → Ger	Ger → NP	NP → Ger
10	66	1512	61	1508	432	829	403	815
20	135	4226	120	4199	972	1997	864	1915
30	165	6391	140	6313	1246	2651	1052	2465
40	178	7935	146	7779	1392	2993	1121	2697
50	186	8920	147	8671	1478	3177	1143	2784
∞	206	10,302	148	9510	1762	3628	1153	2826
	Peak Germany and The Netherlands				Off-peak Germany and The Netherlands			
	Ne → Ger	Ger → Ne	Ne → Ger	Ger → Ne	Ne → Ger	Ger → Ne	Ne → Ger	Ger → Ne
10	816	848	860	788	881	752	858	771
20	2377	2475	2491	2279	2540	2156	2465	2210
30	3842	4011	4006	3659	4062	3432	3932	3515
40	5215	5459	5407	4932	5455	4585	5265	4692
50	6498	6824	6699	6104	6720	5624	6469	5749
∞	22,461	27,929	19,424	17,788	15,895	12,564	14,623	12,551
	Peak Netherlands and UK				Off-peak Netherlands and UK			
	UK → Ne	Ne → UK	UK → Ne	Ne → UK	UK → Ne	Ne → UK	UK → Ne	Ne → UK
10	390	971	347	993	18	1586	14	1582
20	939	2438	803	2471	31	4364	21	4320
30	1283	3313	1054	3321	37	6351	22	6214
40	1501	3770	1187	3732	39	7506	22	7239
50	1641	3996	1254	3909	41	8074	22	7676
∞	2112	4359	1308	4033	44	8606	22	7892

for both on-peak and off-peak segments of the day. Note that these values are for the use of the interconnector during the 365 days of the one-year lease. The total value of the lease is given by the sum of the four options.

Table 5 shows the value of a one-year interconnector lease for both peak and off-peak use. In the first column we show the different assumptions for the value of the liquidity cap M. The second and third columns show values of the flow from one market to the other during peak time when the model includes the seasonal trend $f(T)$, the OU Gaussian shocks $X(T)$, and the jumps $Y(T)$. For example, if $M=10$ Euros/MWh, the value of the 365 options to send 1 MWh of peak electricity from Germany to France (France to Germany) during peak times is 601 Euros (251 Euros), and if the cap is $M=50$ the 365 options are worth 1457 Euros (641 Euros). In the same table, columns 4 and 5, show the value of the interconnector for peak time if the jump intensities are set to zero. In this case, if $M=10$ Euros/MWh, the 365 options to send peak electricity from Germany to France (France to Germany) is 572 Euros (209 Euros), or if the cap is $M=50$ Euros/MWh the value of the options is 1082 Euros (357 Euros).[15] Hence, if we use this example as a proxy for the value added by the jumps, then the difference between columns 2 and 4 (3 and 5) indicates how much value can be extracted from using the interconnector when there are jumps in the spread between Germany and France during peak hours. Columns 6 through 9 can be interpreted in the same way, but they refer to electricity flows during off-peak hours.

From Table 5 we see that the effect of the liquidity cap is different across the markets we study. For example, if the cap between Germany and The Netherlands is reduced from $M=\infty$ to $M=50$ Euros/MWh, the value of the interconnector decreases by almost 75%. If we draw the same comparison in the UK–Netherlands market, the value of the interconnector only decreases by 8%. These different effects of the

[15] These values are calculated employing Eq. (4), where location A is France and location B is Germany.

Table 6
Value of one-year Interconnector lease without including the seasonal component. The model for the spread is Eqs. (11), (12), and (13). The parameters of the seasonal component $f(T)$ are given in Table 2; the parameter estimates of the OU, $X(T)$, are in Panel B of Table 4; and we set all intensities λ of Table 3 to zero. Although the estimation of the parameters is performed including the seasonal component $f(T)$ we value the one-year lease without including the seasonal component $f(T)$. Not including the predictable component $f(T)$ allows us to quantify how much of the value of the interconnector is derived from the deterministic seasonal level by comparing these results with those of Table 5.

$S(T)=$	OU and jumps		OU		OU and jumps		OU	
	$X(T)+Y(T)$		$X(T)$		$X(T)+Y(T)$		$X(T)$	
M	Peak France and Germany				Off-peak France and Germany			
	Ger → Fr	Fr → Ger	Ger → Fr	Fr → Ger	Ger → Fr	Fr → Ger	Ger → Fr	Fr → Ger
10	397	386	352	352	579	596	549	549
20	735	703	579	579	1294	1347	1157	1157
30	862	811	611	611	1642	1729	1376	1376
40	935	869	614	614	1823	1941	1441	1441
50	987	909	614	614	1932	2076	1458	1458
∞	1194	1039	614	614	2290	2613	1463	1463
	Peak France and UK				Off-peak France and UK			
	UK → Fr	Fr → UK	UK → Fr	Fr → UK	UK → Fr	Fr → UK	UK → Fr	Fr → UK
10	649	615	614	614	522	495	481	481
20	1542	1443	1411	1411	1063	993	916	916
30	2039	1885	1798	1798	1267	1170	1024	1024
40	2306	2108	1959	1959	1360	1246	1044	1044
50	2456	2223	2017	2017	1415	1289	1047	1047
∞	2921	2506	2041	2041	1569	1408	1048	1048
	Peak Nord Pool and Germany				Off-peak Nord Pool and Germany			
	Ger → NP	NP → Ger	Ger → NP	NP → Ger	Ger → NP	NP → Ger	Ger → NP	NP → Ger
10	594	663	618	618	608	625	593	593
20	1379	1572	1429	1429	1405	1454	1328	1328
30	1777	2071	1832	1832	1819	1893	1657	1657
40	1957	2330	2005	2005	2030	2124	1785	1785
50	2034	2465	2071	2071	2147	2255	1829	1829
∞	2111	2765	2102	2102	2478	2633	1849	1849
	Peak Germany and The Netherlands				Off-peak Germany and The Netherlands			
	Ne → Ger	Ger → Ne	Ne → Ger	Ger → Ne	Ne → Ger	Ger → Ne	Ne → Ger	Ger → Ne
10	786	878	824	824	840	793	814	814
20	2287	2563	2384	2384	2416	2277	2337	2337
30	3694	4154	3830	3830	3859	3630	3722	3722
40	5011	5655	5165	5165	5173	4857	4975	4975
50	6242	7069	6396	6396	6365	5965	6105	6105
∞	21,622	28,738	18,575	18,575	14,800	13,544	13,559	13,559
	Peak Netherlands and UK				Off-peak Netherlands and UK			
	UK → Ne	Ne → UK	UK → Ne	Ne → UK	UK → Ne	Ne → UK	UK → Ne	Ne → UK
10	641	609	610	610	473	578	507	507
20	1520	1427	1403	1403	944	1219	1006	1006
30	2024	1875	1809	1809	1088	1475	1151	1151
40	2316	2119	2006	2006	1128	1583	1185	1185
50	2495	2255	2101	2101	1141	1637	1191	1191
∞	3042	2528	2178	2178	1153	1727	1193	1193

liquidity cap are due to the particular characteristics of the spread in each market: seasonal component, volatility of the OU process, jump intensities and jump sizes. For example, the cap has an important effect on the markets where the spread undergoes frequent and large jumps which is the case for Germany–The Netherlands. And in general, depending on the depth of the market, the results of Table 5 show that the jumps in the spread can account for between 1% and 40% of the total value of the interconnector.

Moreover, Table 5 shows that the peak and off-peak value of interconnection can also be different even within the same market, hence there are markets where most of the value of the interconnector is derived from flows in one direction and for a particular segment of the day. For instance, most of the value of the interconnection between The Netherlands and the United Kingdom is provided by off-peak hours, while most of the value between Germany and Nord Pool is during peak hours.

In Table 6 we investigate the case when the seasonal trend is set to zero for all T. The table gives an indication of what happens to the value of the interconnector if it is assumed that for future dates the parameters of the OU and jump component are the same, but that the deterministic seasonal trend of the spread between the locations is zero. By comparing these values to those of Table 5, we can see that the seasonal trend adds considerable value to the interconnector, especially for cases with high liquidity cap M. For example, if the spread is modeled with $f(T)+X(T)+Y(T)$, then the value of an interconnector between France and Germany is 6252 Euros ($M=50$ Euros/MWh), but if we assume that the seasonal function is set to zero, $f(T)=0$, then the value drops to 5904 Euros.[16] Moreover, Table 6 also shows that the cases in

[16] These figures are obtained from adding up the values of the peak and off-peak of the flows between Germany and France. Note that the lessee owns the four options: Germany to France, France to Germany during peak time as well as off-peak time.

Table 5 where the interconnector was mostly a uni-directional flow are mainly due to the seasonal component.

Finally, based on the statistics and estimates in Tables 1–4, we provide some rules of thumb that help to interpret and provide a qualitative summary of the results in Tables 5 and 6. First, as in financial options, high volatilities in the underlying spread means higher interconnector values. This feature is clearly exhibited by the peak-hour interconnections in Germany–Netherlands, Netherlands–UK, and Germany–Nordpool, where the standard deviations of the spread of the peak-hours are the largest and so are their interconnector values (see summary statistics in Table 1). As discussed above, the variance of the spread process depends on four elements: the variance of the diffusion process, the size and arrival intensity of jumps, the mean-reversion rate of the diffusion, and the mean reversion rate of the jumps. Each one of these elements affects the value of the interconnector.

Similar to the effect of volatility of the underlying asset of financial options, the higher the volatility of the diffusion part of the spread, the higher is the value of the interconnector. By inspecting Table 6 (columns 4 and 5 for peak hours, and 8 and 9 for off-peak hours) where we turn off the effect of the seasonal trend and the jumps in the spread, we can see how the diffusion component of the spread model affects the value of the interconnector. For example, if we look at the peak-hour and the off-peak-hour spreads of Germany–Nordpool we see that they have similar mean-reversion rates, but the volatility of the diffusion component of the peak spread is slightly higher than that of the off-peak spread. Thus, as expected, the interconnector values for peak hours are higher than those for off-peak hours.

The largest values of the interconnector correspond to markets where the size and frequency of jumps is significantly large (e.g.Germany–The Netherlands peak hours in Table 5). The effect of the jump component

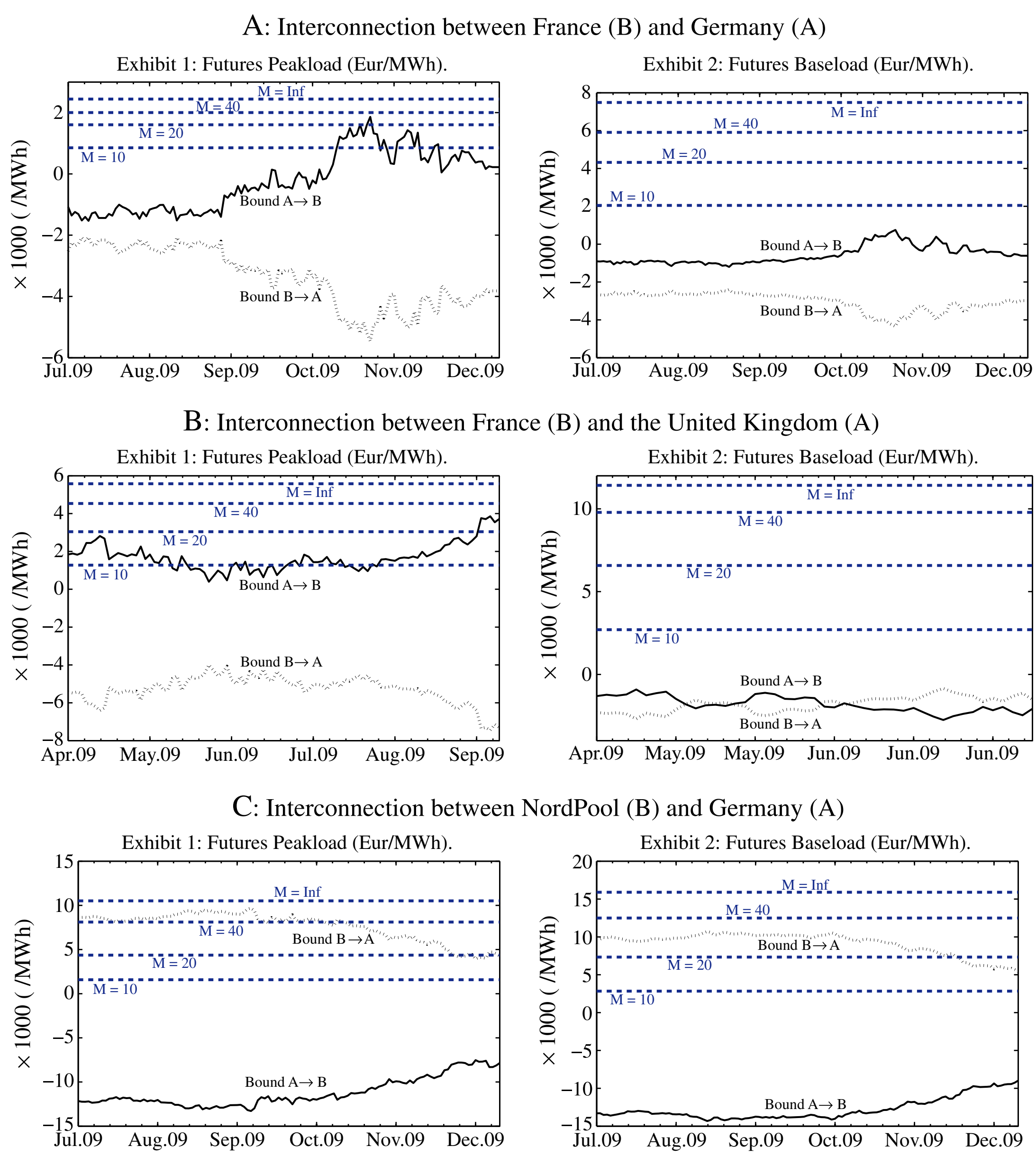

Fig. 3. Empirical no-arbitrage bounds. Dashed blue lines show the corresponding interconnector value of Table 5 for different liquidity constraints $M = 10, 20, 40$ and ∞. Black lines show the bounds computed from Eqs. (9) and (10) for peak hours and the respective ones for off-peak hours. The no-arbitrage bounds resulting from buying a forward contract in location B, and shorting a forward contract for location A (bound A→B) are depicted by a solid line. Similarly, the dotted line represents the bound resulting from being long a forward contract in location A, and short a forward in location B (bound B→A).

depends on the depth of the market; increases in the liquidity cap M imply significant increases in the value of the interconnectors.

At the same time, for similar levels of variance in the diffusion process of the spread, a larger rate of mean reversion of the diffusion component decreases the value of the interconnector. An example of this effect can be found in the results for the France–Germany interconnector values when the jump component is "switched off" (see Table 6, columns 4 and 5 for peak hours, and 8 and 9 for off-peak hours). There we observe that the off-peak interconnector values are larger than the corresponding peak interconnector values while the diffusion volatilities of peak and off-peak spreads are similar. Thus, this difference in the option values can be attributed mainly to the difference in the speed of mean reversion between the diffusion processes of the peak and off-peak spreads.

If we take into account the jump component, we also observe that, when the arrival intensities and jump sizes are similar, the speeds of mean reversion of the jump and diffusive component, become important drivers of the interconnector value. Thus, spreads with higher mean reversion rates of the jump processes have the lowest interconnector value.

8.4. No-arbitrage bounds

The values in Tables 5 and 6 do not take into account the arbitrage bounds that forward or futures prices could impose on the valuation of the interconnector. In this subsection we use the prices shown in Table 5 for the benchmark model $S(T)=f(T)+X(T)+Y(T)$ in columns 2, 3 (peak) and columns 6, 7 (off-peak) as a reference. For a fixed liquidity cap M, the no-arbitrage bounds require that the sum of the values in columns 2 and 3 obey the peak electricity bounds (9) and (10). Similarly, for a given cap M the sum of values in columns 6 and 7 must obey the no-arbitrage bounds for off-peak hours.

To verify whether the prices reported in the valuation tables are arbitrage-free, we look at a particular example. Recall that the values reported in the tables are for a one-year lease between January 1, 2010 and December 31, 2010 and that these values were calculated employing data that ended in June 2009 (below we show in detail how the bound for peak time on October 29, 2009 between Germany and France is calculated). The no-arbitrage bounds will depend on the market data on the day the calculation is performed; here it can be any day between July 1, 2009 and December 31, 2009. Fig. 3 shows the bounds for every day during this period for France and Germany, France and UK, and Nord Pool and Germany. We can observe that the bounds exhibit considerable variation over time. These pronounced changes in the bounds are a reflection of changes in futures prices due to changes in: market expectations, fuel prices, changes in risk-premia, weather predictions, etc.

For example, the no-arbitrage bound for peak electricity between France and Germany on October 29, 2009 is calculated in the following way. Assume that $M=20$, and the price for a one-year lease of 1 MWh capacity, for a one-hour slot during the peak time between Germany and France, is 1603 Euros (1136 Euros for transmission from Germany to France plus 467 Euros for transmission from France to Germany; see Table 5). On October 29, 2009, yearly peak-load futures (delivering electricity during peak time throughout 2010) are trading at 77.63 Euros/MWh in France and 68.15 Euros/MWh in Germany. Now, consider the following strategy: 1) Purchase the yearly peak-load forward in Germany and sell a yearly peak-load forward in France; 2) Purchase interconnector capacity for 2010, during peak time at 1603 Euros (4.40 Euros/MWh during 365 days); 3) Every day during 2010, transmit 1 MWh from Germany to France and pay transmission costs of 5 Euros/MWh. Following Eq. (9), the no-arbitrage lower bound is 1611 Euros, and the net present value of this arbitrage strategy is 8 Euros (i.e. the difference between the no-arbitrage bound and the value of the interconnector capacity).[17] Therefore, the valuation in Table 5 does not satisfy the no-arbitrage bound and if the lessor is selling capacity on October 29, 2009, he should increase the price from 1603 Euros to at least 1611. However, if the same arbitrage strategy is considered on October 30, 2009, the non-arbitrage bound is equal to 1236 Euros, which is below the value charged by the lessor, and there are no opportunities to arbitrage the price. Exhibit 1 of Panel A in Fig. 3 illustrates this example.

9. Conclusions

In this paper we show how to value an electricity interconnector as an asset that gives the owner the optionality to manage electricity flows between two markets. In financial terms, the value of the interconnector is the same as a strip of real options written on the spread between the power prices of two markets. We model the spread prices based on a: seasonal trend, mean-reverting Gaussian process, and mean-reverting jump process.

As a first contribution, we express the value of these real options in closed-form in the presence of mean-reverting jumps processes. We also propose a methodology to detect jumps in the spread that addresses the possible miss-classification of mean reversion as jumps. We estimate the parameters of the spread model and find that the introduction of jumps in the model delivers gains in the in-sample performance of between 20% and 48% with respect to a misspecified or "naive" modeling which jumps are not included.

We value a hypothetical one-year lease of the interconnector between five pairs of European neighboring markets. The total value of the lease is the sum of the premia of 1460 (4 times 365) real options on the spread: 365 options to transmit electricity from market A to B during on-peak hours; 365 to transmit electricity from A to B during off-peak hours; and the same number for transmitting both on- and off-peak power from market B to A. We show valuations under different assumptions about the seasonal component of the spread and different liquidity caps, which proxy for the depth of the interconnected power markets. Finally, we provide some rules of thumb to interpret the different interconnector values.

Although we cast the problem in terms of real options, where the statistical distribution of the spread and the risk-adjusted discount rate are key ingredients in the valuation, we also derive no-arbitrage lower bounds for the value of the interconnector in terms of electricity futures contracts. We show that most of the time these bounds are satisfied, but there are days where the value of the interconnector is given by the no-arbitrage bound instead of the price delivered by the sum of the prices of real options.

We find that the seasonal component is an important factor that determines the direction of the interconnector value. We also show that the variance of the diffusion process, the size of the jumps, and the mean-reversion rates of the diffusion and jump processes are key elements to determine the final value of the interconnector. Some of our valuation findings indicate that, depending on the depth of the market, the jumps in the spread can account for between 1% and 40% of the total value of the interconnector. The presence of a liquidity constraint reduces the value of interconnectors specially for those interconnected markets where the jump component plays a key role in the variance of the spread price. The two markets where an interconnector would be most (least) valuable is between Germany and The Netherlands (between France and Germany). The markets where off-peak transmission between the two countries is more valuable than transmission during the peak times are: France and Germany, France and UK, and The Netherlands and UK.

[17] That is, $\sum_{t=1}^{365} e^{-rt}$ $(77.63-68.15-5)=1611$ Euros/MWh; the risk-free rate employed in the calculations is 3%. Note that this profit is for 1 MWh but could be much more if we consider that an interconnector may have a capacity of 1000 MWh (or more) and that there are 12 peak hourly slots every day. In this case the total profit is 96,000 Euros.

Appendix. Proofs

A. Prices of Call Options on the Spread

To calculate European option prices using transform techniques we need the Fourier transform of the payoff of the option and the characteristic function (cf) of the process for the spread. We start from the result that for a Lévy process $L(t)$ and a deterministic function $h(t)$

$$\mathbb{E}\left[e^{i\xi\int_h^T h(u)dL_u}\right] = e^{\int_h^T \Psi(h(u)\xi)du}, \quad \xi=\xi_r + i\xi_i \text{ and } \xi_r, \xi_i \in \mathbb{R}.$$

Hence, for a compound Poisson process with time-dependent intensity $\lambda(t)$ and $h(u) = e^{-\beta(T-u)}$

$$\mathbb{E}\left[e^{i\xi\int_h^T h(u)JdN_u}\right] = e^{\int_t^T \left(\Psi_J\left(e^{-\beta(T-u)}\xi\right)-1\right)\lambda(u)du}$$

where $\Psi_J(\xi)$ is the cf of the jump random variable J.

In the particular case where jump sizes are exponentially distributed the cfs are

$$\mathbb{E}\left[e^{i\xi J}\right] = \frac{\eta_1}{\eta_1 - i\xi} \text{ with } \xi_i > -\eta_1 \text{ and } \mathbb{E}\left[e^{i\xi J}\right] = \frac{\eta_2}{\eta_2 + i\xi} \text{ with } \xi_i > \eta_2.$$

Since we assume that the intensity parameters of the inhomogeneous Poisson processes are either constant or piecewise constant, we must calculate the integrals $\int_t^T \Psi_J(e^{-\beta(T-u)}\xi)du$, over a time interval $(t,T]$, for the positive and negative jumps, respectively:

$$\int_t^T \left(\frac{\eta_1}{\eta_1 - i\xi e^{-\beta(T-u)}} - 1\right)du = \frac{1}{\beta}\ln\left(\frac{i\xi e^{-\beta(T-t)} - \eta_1}{i\xi - \eta_1}\right),$$

and

$$\int_t^T \left(\frac{\eta_2}{\eta_2 + i\xi e^{-\beta(T-u)}} - 1\right)du = \frac{1}{\beta}\ln\left(\frac{i\xi e^{-\beta(T-t)} + \eta_2}{i\xi + \eta_2}\right).$$

Thus, the cf of $S^{A,B}(T)$, conditioned on information up until time t, and assuming constant intensity parameters for the jump processes between $(t,T]$, is given by

$$\Psi_S^{A,B}(\xi) = \mathbb{E}\left[e^{i\xi S(T)}\right] = e^{i\xi h(T) - \frac{1}{2}\xi^2\int_t^T e^{-2\alpha(T-u)}\sigma^2(u)du}\left(\frac{i\xi e^{-\beta(T-t)} - \eta_1}{i\xi - \eta_1}\right)^{\frac{\lambda^+}{\beta}} \left(\frac{\eta_2 + i\xi e^{-\beta(T-t)}}{\eta_2 + i\xi}\right)^{\frac{\lambda^-}{\beta}} \quad (26)$$

where $h(T) = f(T) + X(t)e^{-\alpha(T-t)} + Y(t)e^{-\beta(T-t)}, -\eta_1 < \xi_i < \eta_2$, and λ^+ and λ^- are the intensities of the Poisson arrival of positive and negative jumps respectively.

Therefore, the price of European options on the spread $S^{A,B}(T)$, with expiry T and strike $K^{A,B}$ is given by an integral along a straight line in $\mathbb{C}$ parallel to $\mathbb{R}$ with $-\eta_2 < \xi_i < \eta_1$,

$$C^{A,B}\left(S, \infty, t; T, K^{A,B}\right) = \frac{e^{-r(T-t)}}{2\pi}\int_{-\infty + i\xi_i}^{\infty + i\xi_i} \Psi_S(-\xi)\Pi^{A,B}(\xi)d\xi \quad (27)$$

and $\Pi^{A,B}(\xi)$ is the Fourier transform of the call option payoff:

$$\Pi^{A,B}(\xi) = \int_{-\infty}^{\infty} e^{i\xi x} \max\left(x - K^{A,B}, 0\right)dx = -\frac{e^{i\xi K^{A,B}}}{\xi^2} \quad (28)$$

for $\xi_i > 0$. $\Pi^{B,A}(\xi)$ is obtained in the same way.

Proof of Eq. (18)

Let $Y = S^{A,B}(T)$ and let $f(y)$ the pdf of the normal random variable Y and recall that $\mu(t,T)$ and $v^2(t,T)$ are as in (19). Then $C^{A,B}(S^{A,B}, M, t; T, K^{A,B})$

$$\begin{aligned}
&= e^{-\rho(T-t)}\left\{\int_{K^{A,B}}^{\infty} Yf(y)dy - K^{A,B}\int_{\beta_1}^{\infty}\phi(x)dx - \int_M^{\infty} Yf(y)dy - M\int_{\beta_2}^{\infty}\phi(x)dx\right\} \\
&= e^{-\rho(T-t)}\left\{\int_{K^{A,B}}^{\infty} Yf(y)dy - K^{A,B}\Phi(-\beta_1) - \int_{K^{A,B}}^{\infty} Yf(y)dy - M\Phi(-\beta_2)\right\} \\
&= e^{-\rho(T-t)}\left\{\left[\mu(t,T) - K^{A,B} + v(t,T)\frac{\phi(\beta_1)}{1-\Phi(\beta_1)}\right]\Phi(-\beta_1)\right. \\
&\qquad \left. - \left[\mu(t,T) - M + v(t,T)\frac{\phi(\beta_2)}{1-\Phi(\beta_2)}\right]\Phi(-\beta_2)\right\},
\end{aligned} \quad (29)$$

where $\beta_1 = (K^{A,B} - \mu(t,T))/v(t,T), \beta_2 = (M - \mu(t,T))/v(t,T)$.

Moreover, note that to evaluate the integrals that are of the form

$$\int_{K^{A,B}}^{\infty} yf(y)dy, \quad (30)$$

in Eq. (29), where $f(y)$ and $F(Y)$ are the pdf and cdf of the random variable $Y \sim N(\mu(t,T), v^2(t,T))$, we use Bayes' theorem to obtain

$$\begin{aligned}
\int_{K^{A,B}}^{\infty} yf\left(y|y > K^{A,B}\right)dy &= \int_{K^{A,B}}^{\infty} y\frac{f(y)}{\left(1 - F(K^{A,B})\right)}dy \\
&= \mu(t,T) + v(t,T)\frac{\phi\left(\left(K^{A,B} - \mu(t,T)\right)/v(t,T)\right)}{1 - \Phi\left(\left(K^{A,B} - \mu(t,T)\right)/v(t,T)\right)},
\end{aligned}$$

and for the last equality we use the fact that for a constant K

$$\mathbb{E}[Y|K < Y] = \mu(t,T) + v(t,T)\frac{\phi((K - \mu(t,T))/v(t,T))}{1 - \Phi((K - \mu(t,T))/v(t,T))} \quad (31)$$

and

$$\mathbb{E}[Y|Y < K] = \mu(t,T) - v(t,T)\frac{\phi((K - \mu(t,T))/v(t,T))}{\Phi((K - \mu(t,T))/v(t,T))}. \quad (32)$$

Therefore, the truncated integral

$$\begin{aligned}
\int_{K^{A,B}}^{\infty} Yf(y)dy &= \mu(t,T)\left(1 - \Phi\left(\left(K^{A,B} - \mu(t,T)\right)/v(t,T)\right)\right) \\
&+ v(t,T)\frac{\phi\left(\left(K^{A,B} - \mu(t,T)\right)/v(t,T)\right)}{1 - \Phi\left(\left(K^{A,B} - \mu(t,T)\right)/v(t,T)\right)}\left(1 - \Phi\left(\left(K^{A,B} - \mu(t,T)\right)/v(t,T)\right)\right).
\end{aligned}$$

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